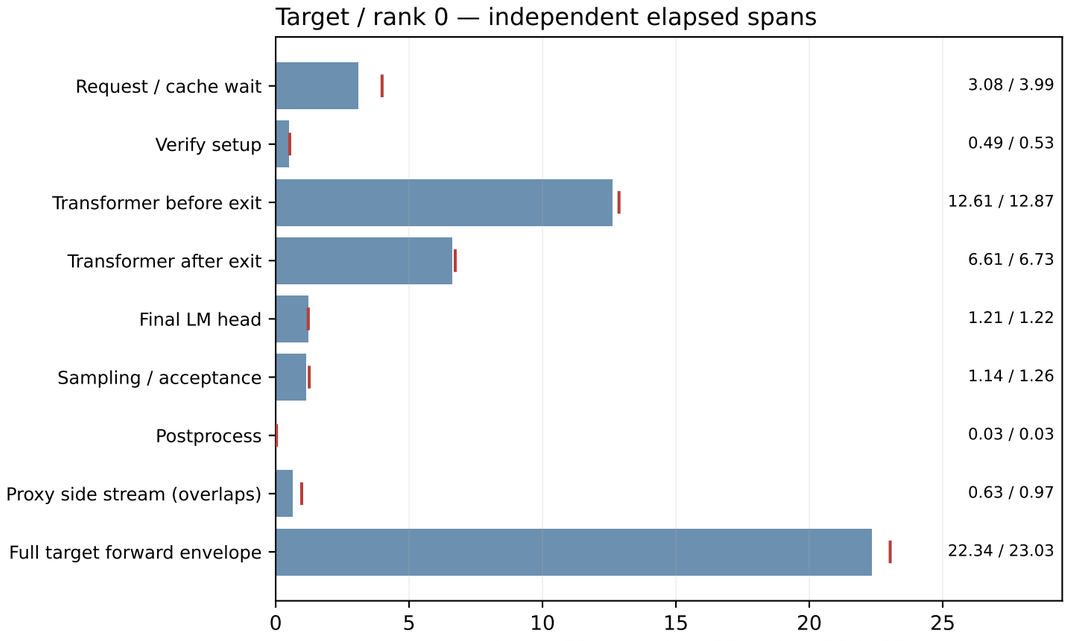
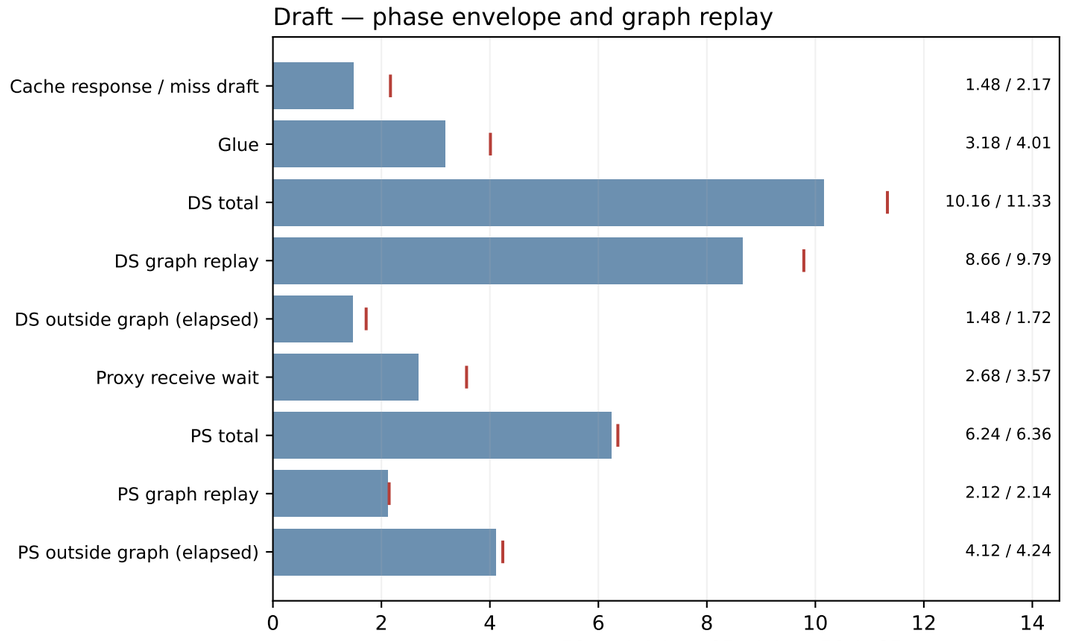
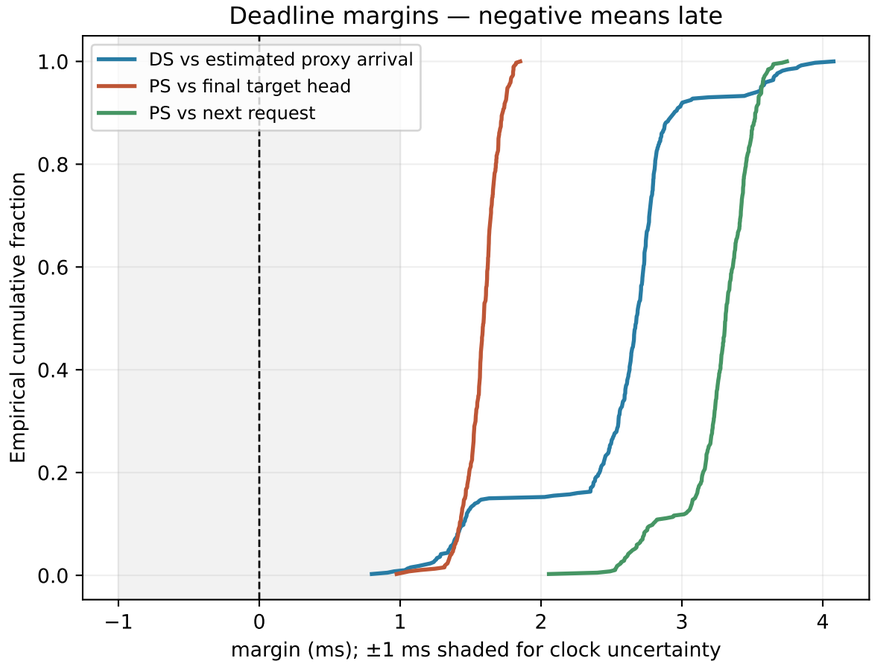
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Component latency, phase deadlines, and every measured step

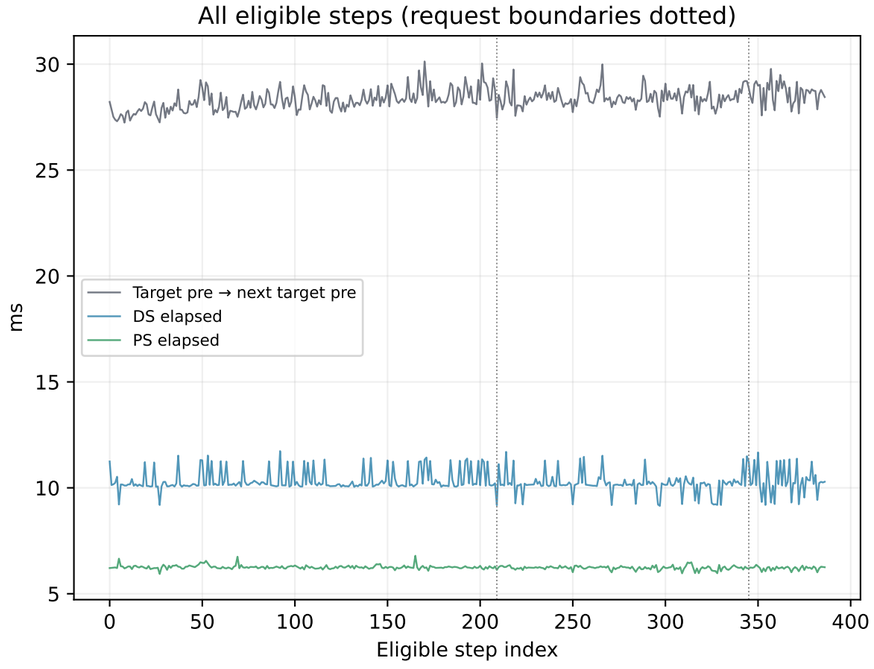
exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256







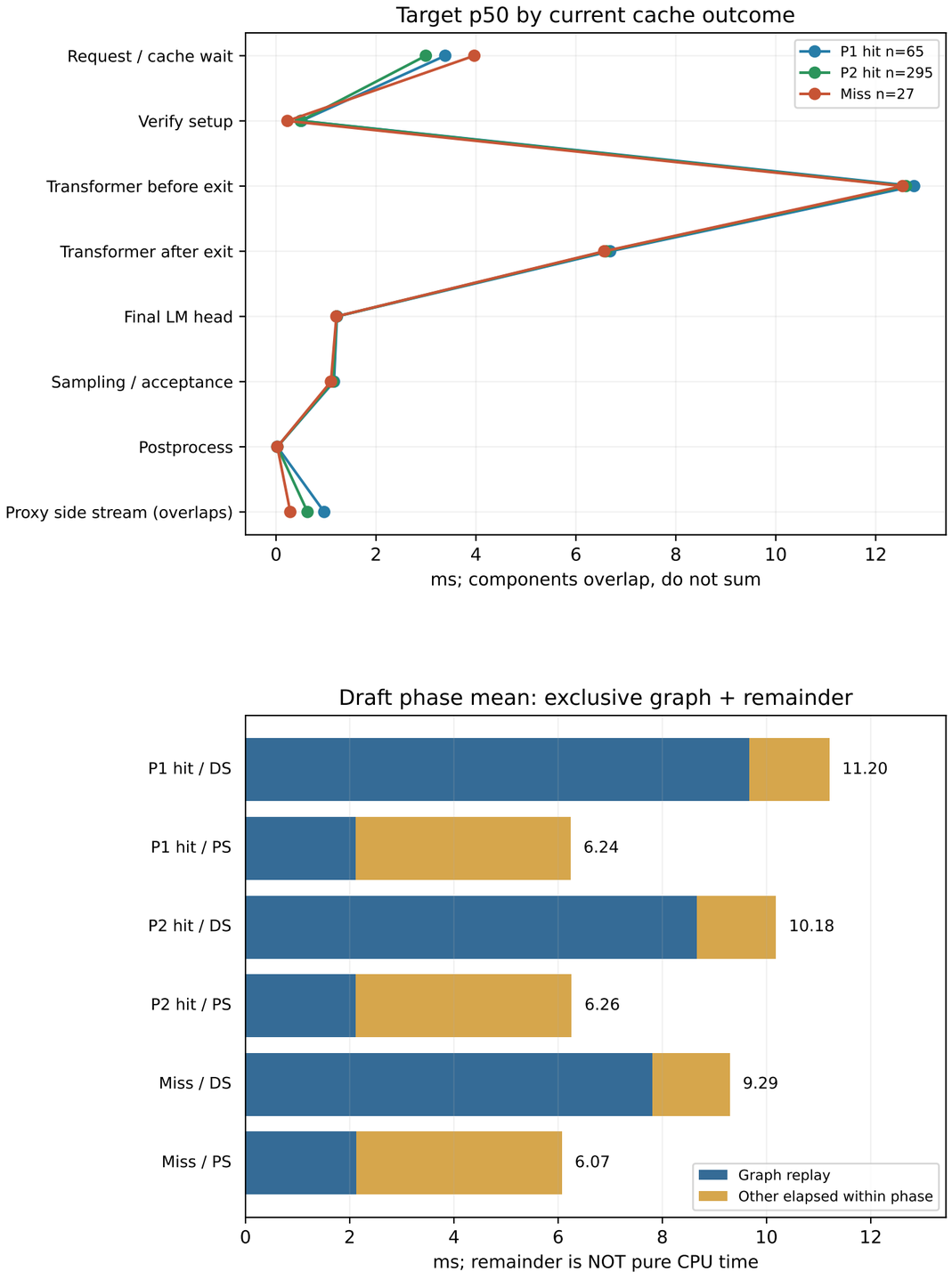


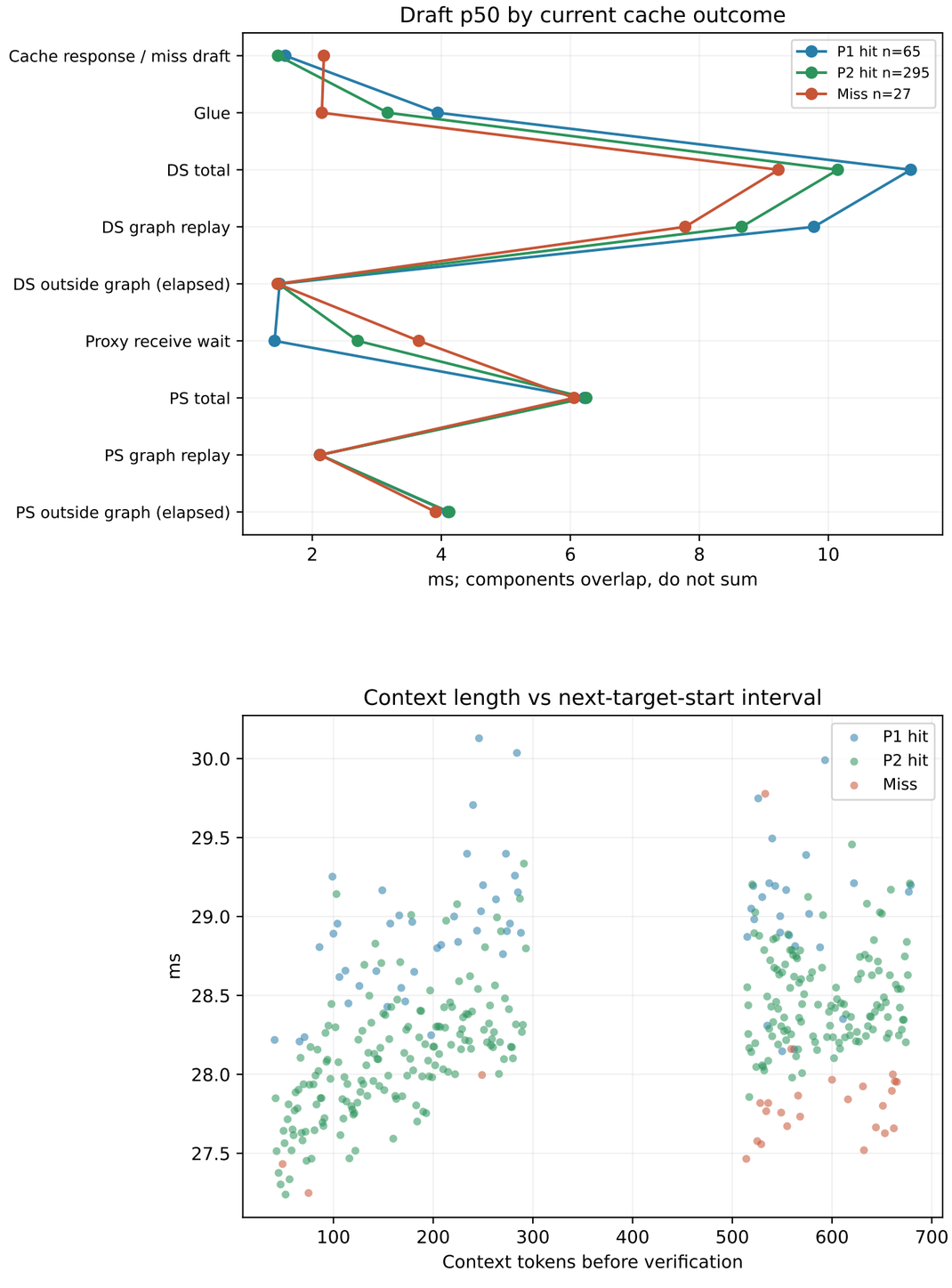
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P1 hit / P2 hit / miss — component distributions and phase composition

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256





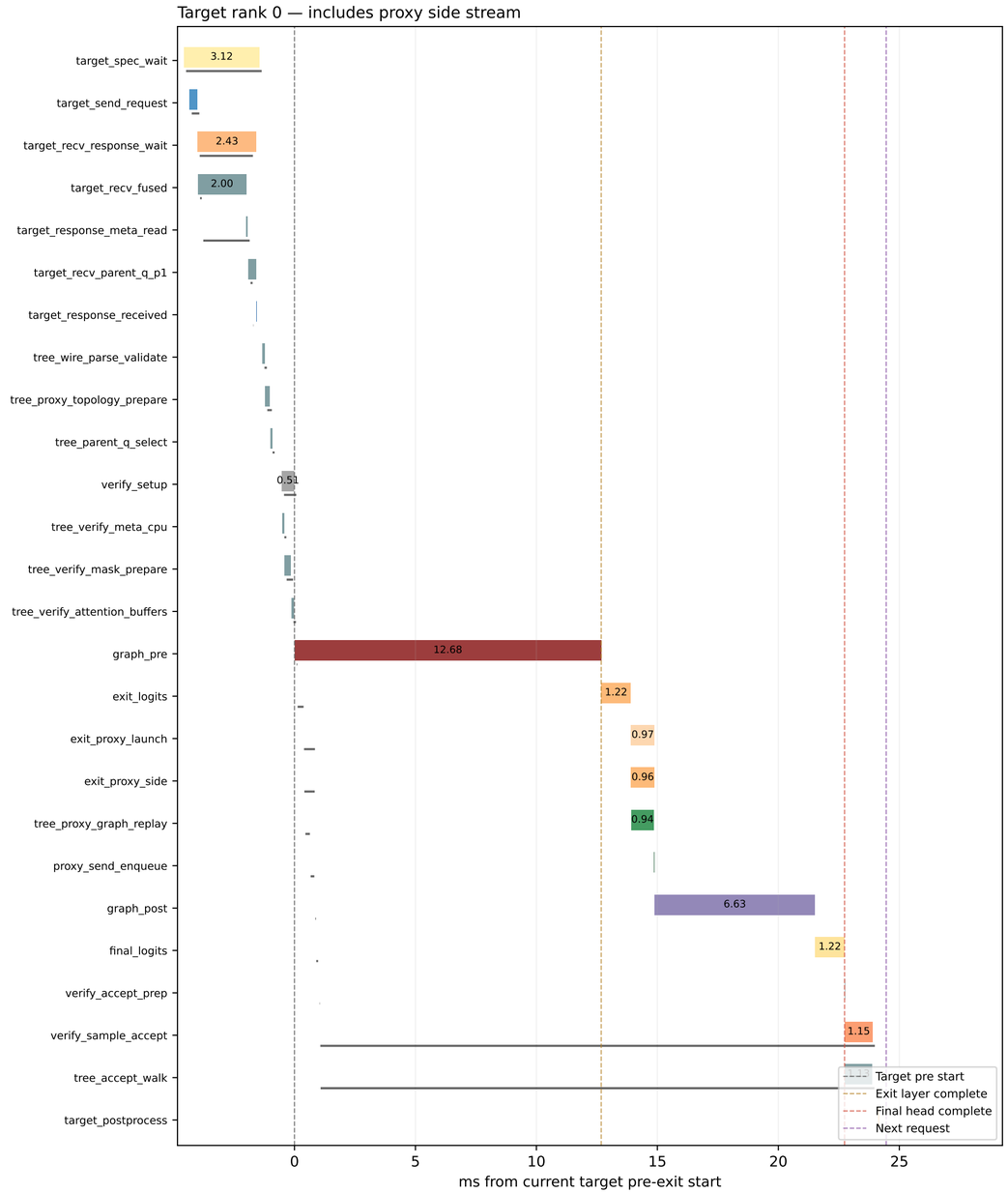
CUDA-event elapsed spans include queue/host stalls; not GPU busy time. Nested/overlapping components are NOT additive.

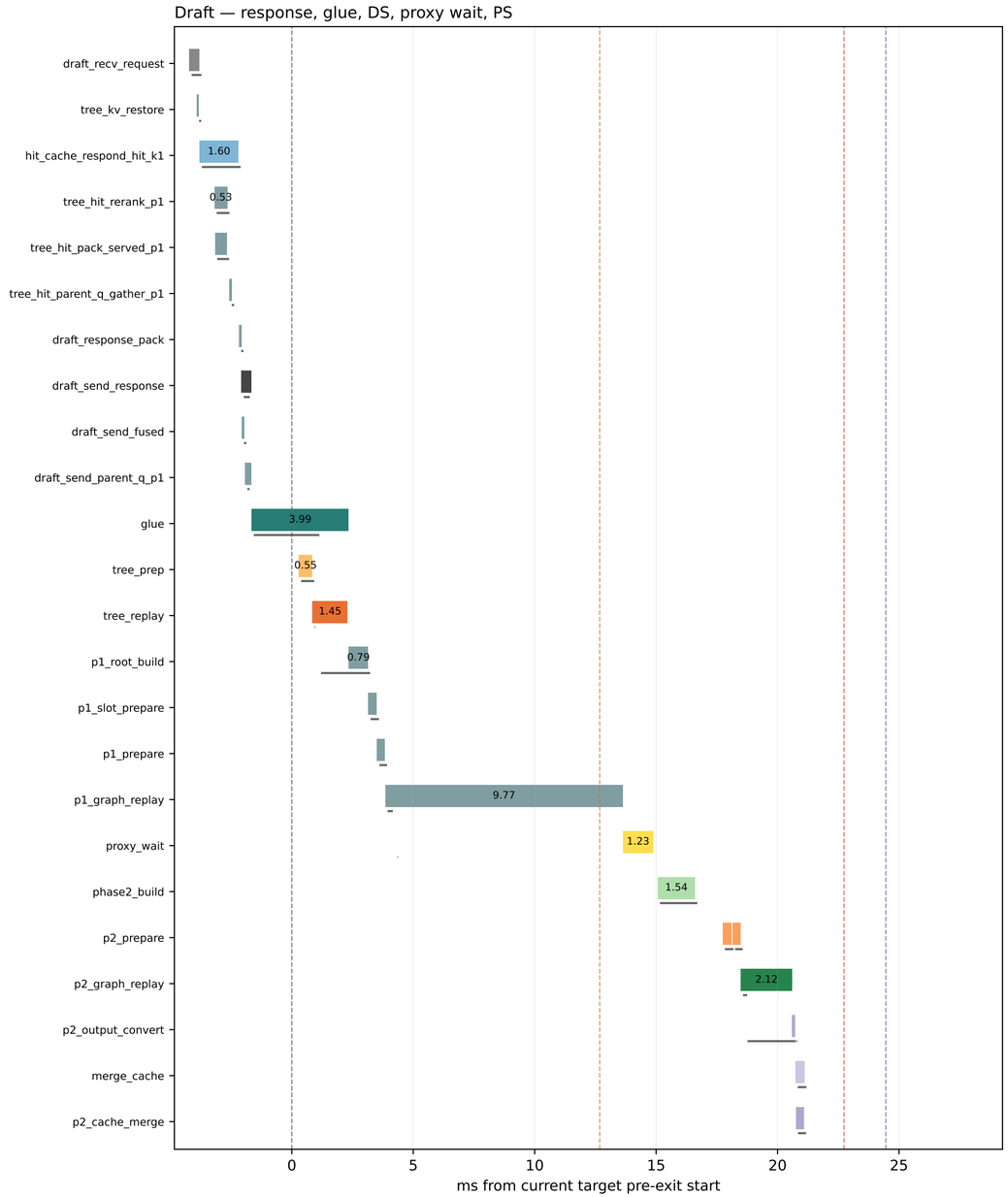
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P1 hit / p25 target-step duration / request 1, UID 84_t0, step 113

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=1; DS margin +1.238 ms (estimated); PS margin +1.641 ms; next outcome P2 hit





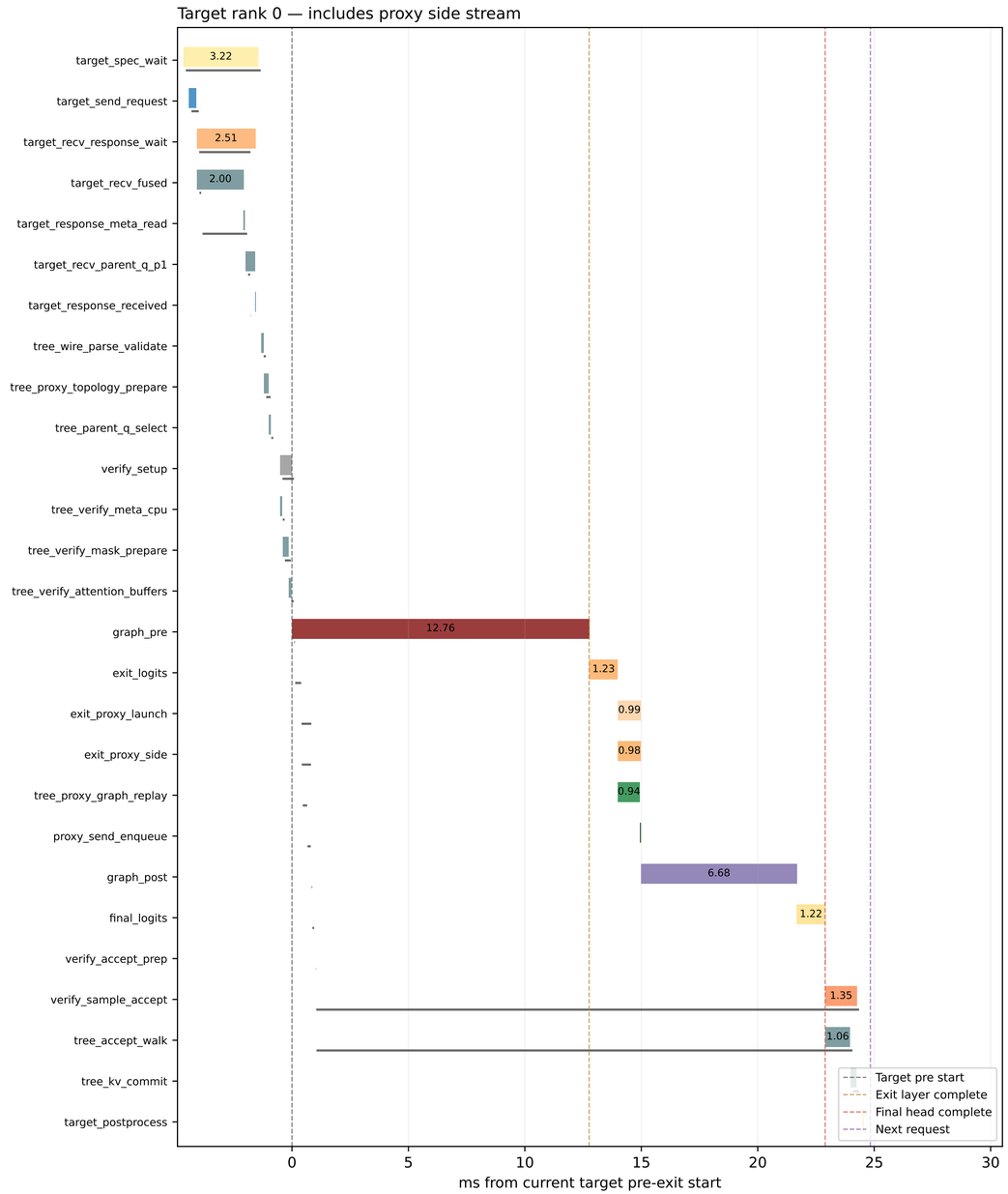
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

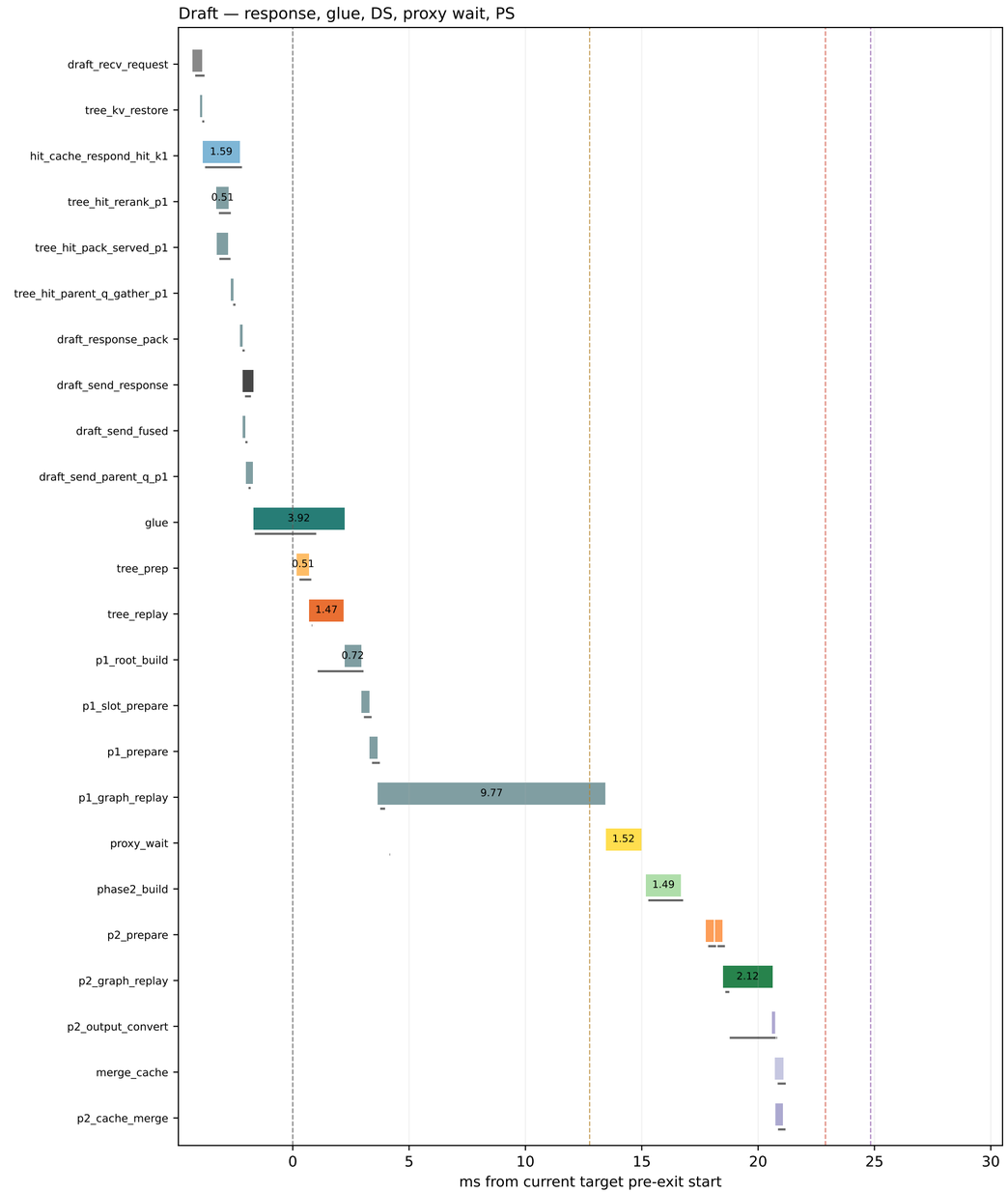
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P1 hit / p50 target-step duration / request 1, UID 84_t0, step 170

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=4; DS margin +1.522 ms (estimated); PS margin +1.804 ms; next outcome P1 hit





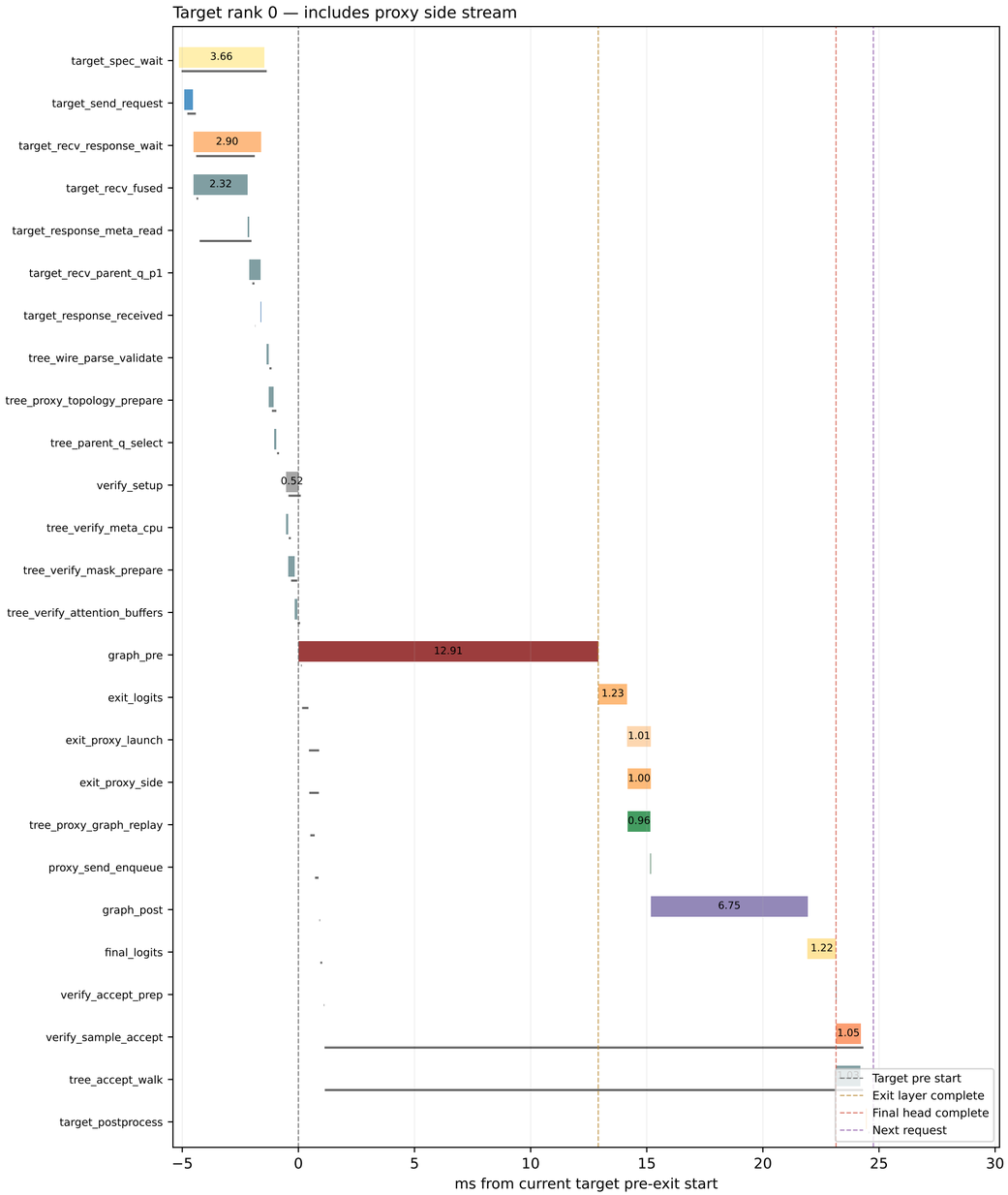
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

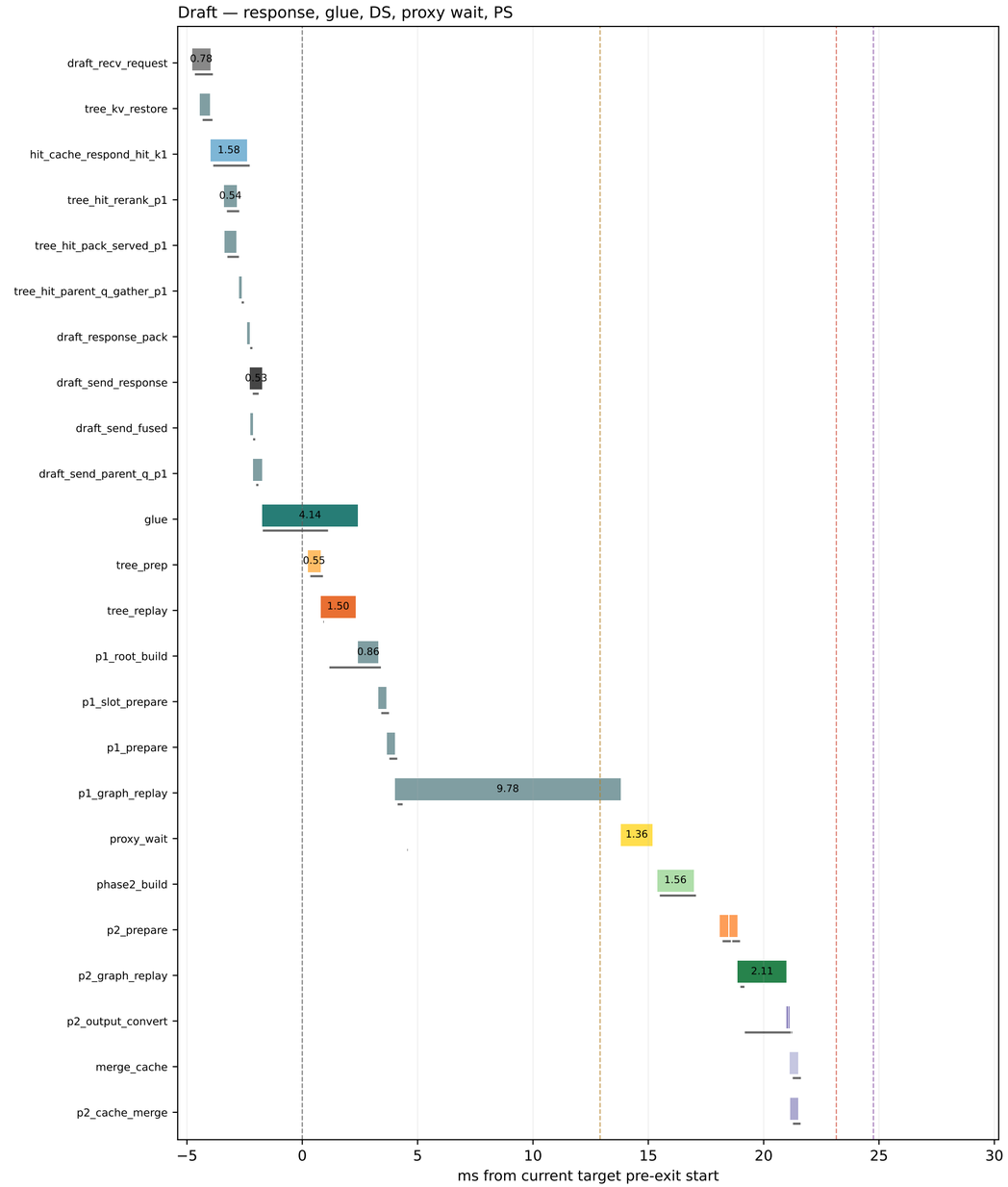
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P1 hit / p75 target-step duration / request 2, UID 244_t0, step 48

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=8, AL incl. recovery=2; DS margin +1.356 ms (estimated); PS margin +1.650 ms; next outcome P2 hit





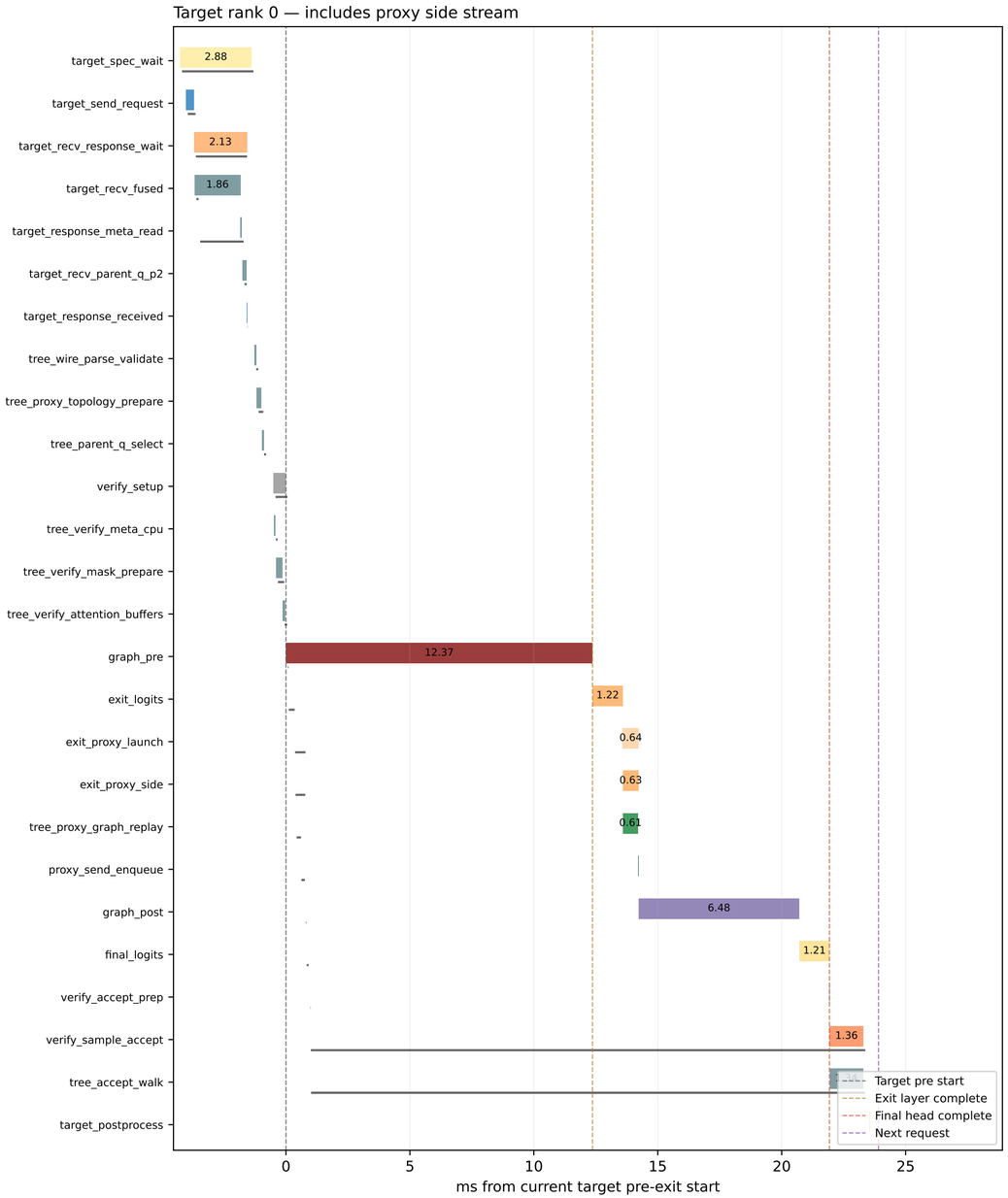
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

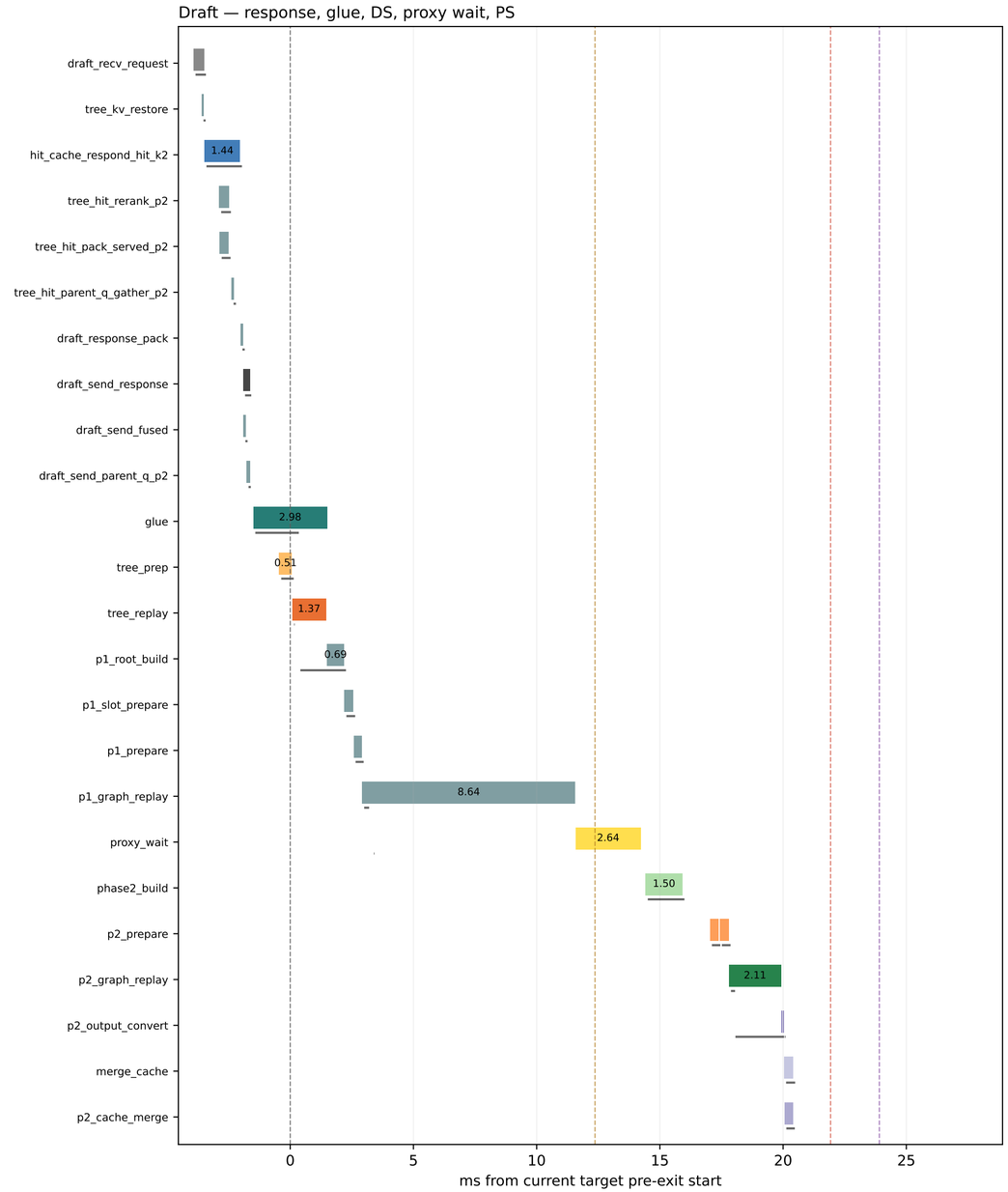
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P2 hit / p25 target-step duration / request 1, UID 84_t0, step 23

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +2.645 ms (estimated); PS margin +1.517 ms; next outcome P2 hit





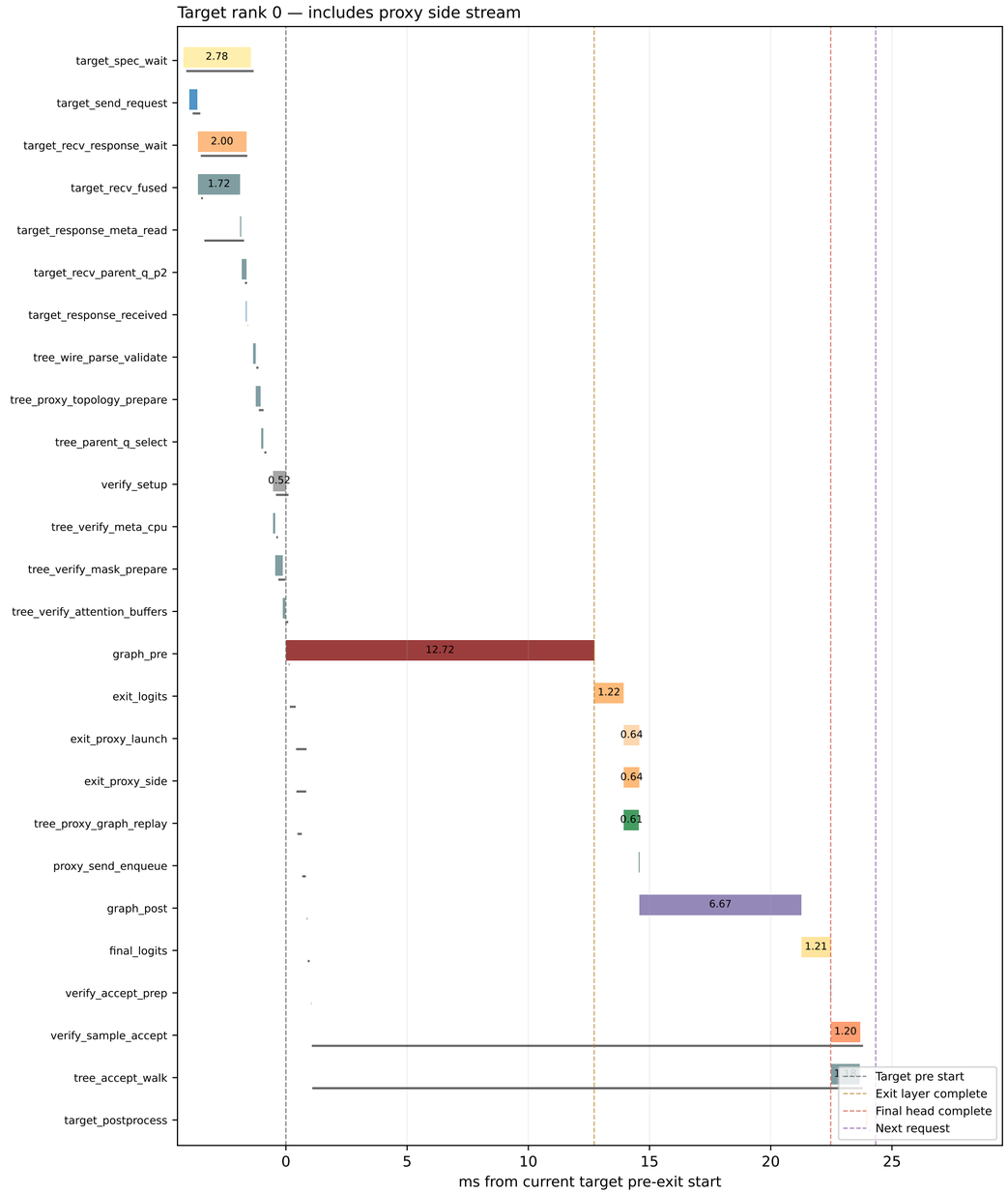
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

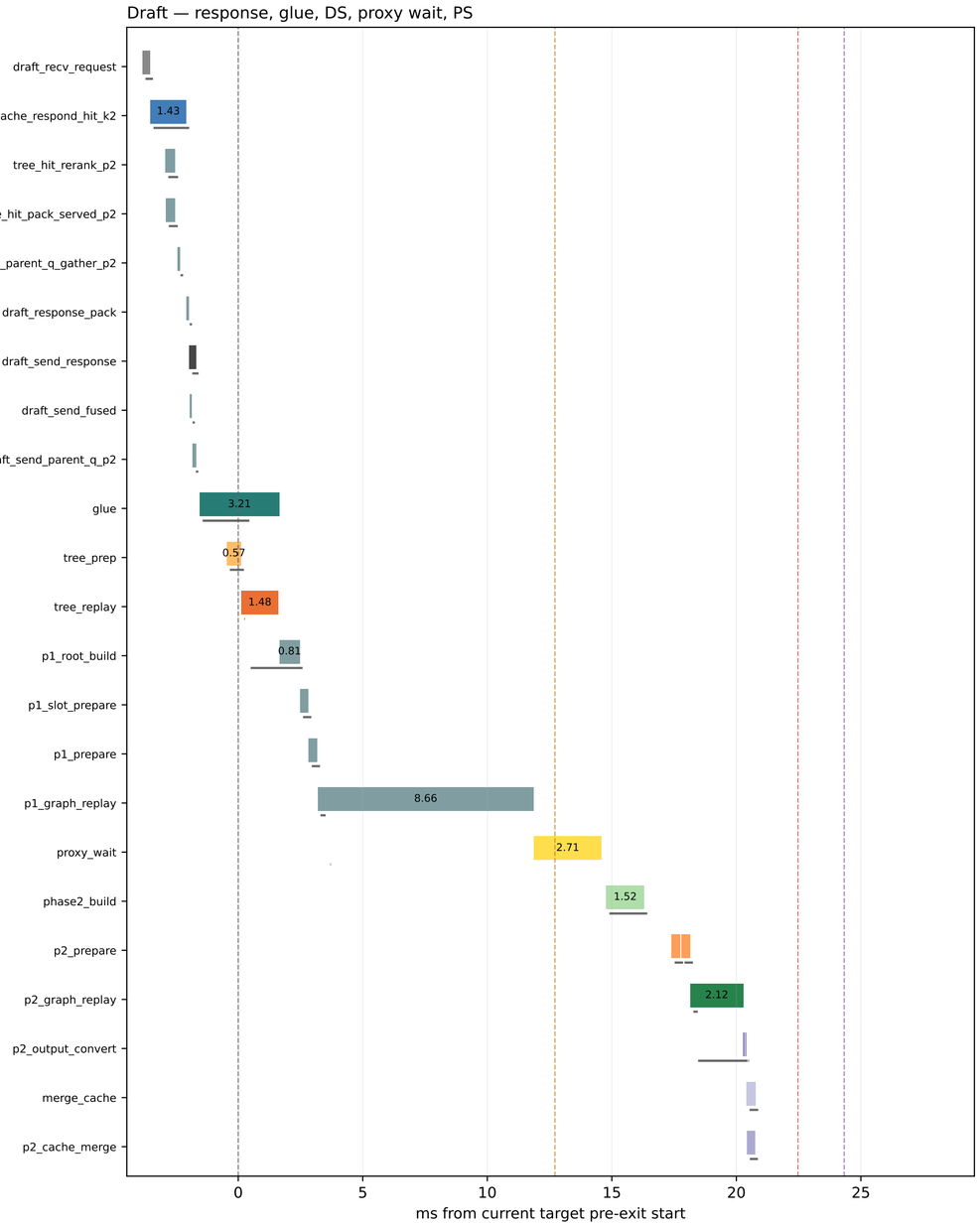
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P2 hit / p50 target-step duration / request 2, UID 244_t0, step 92

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +2.710 ms (estimated); PS margin +1.708 ms; next outcome P2 hit





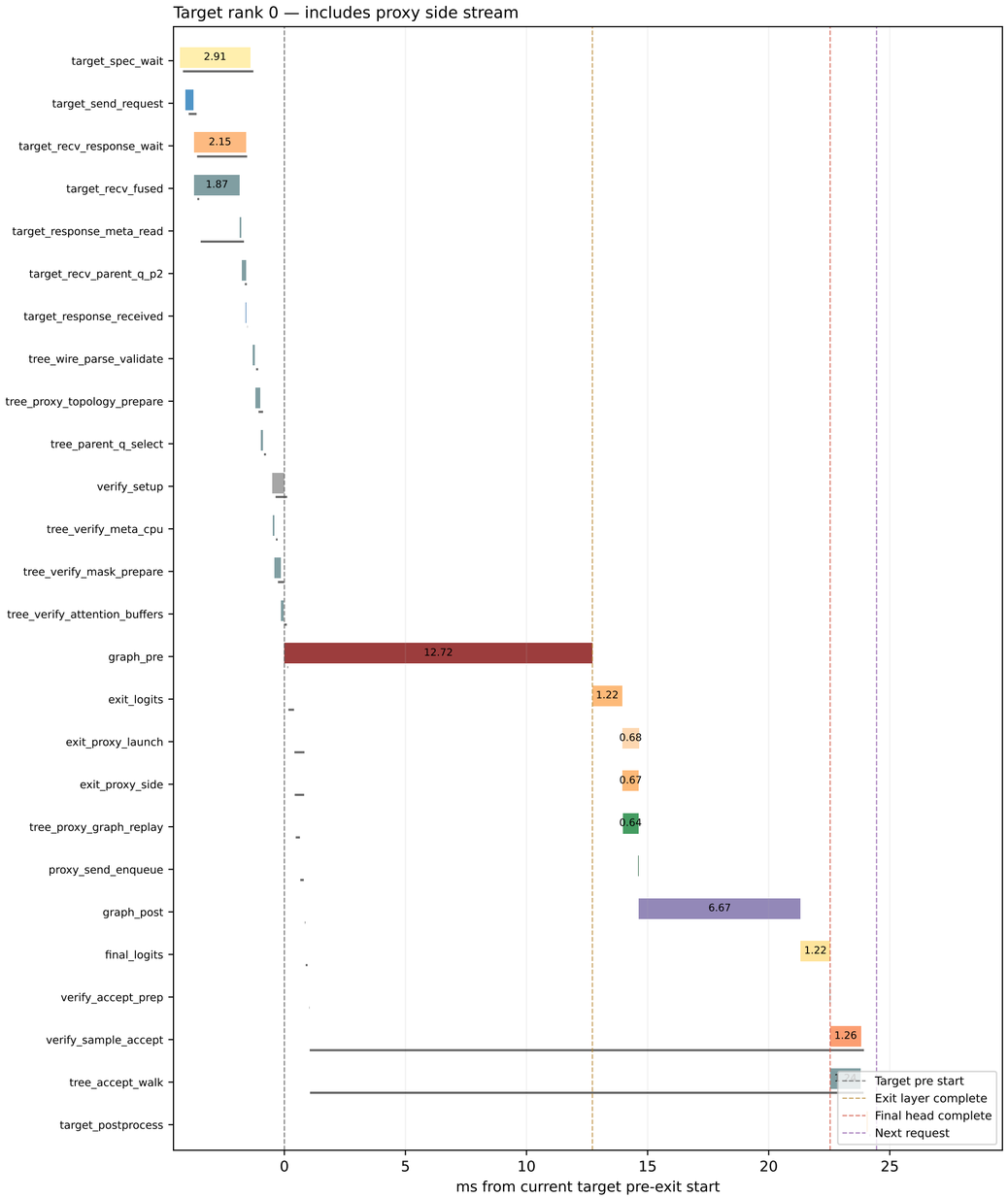
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

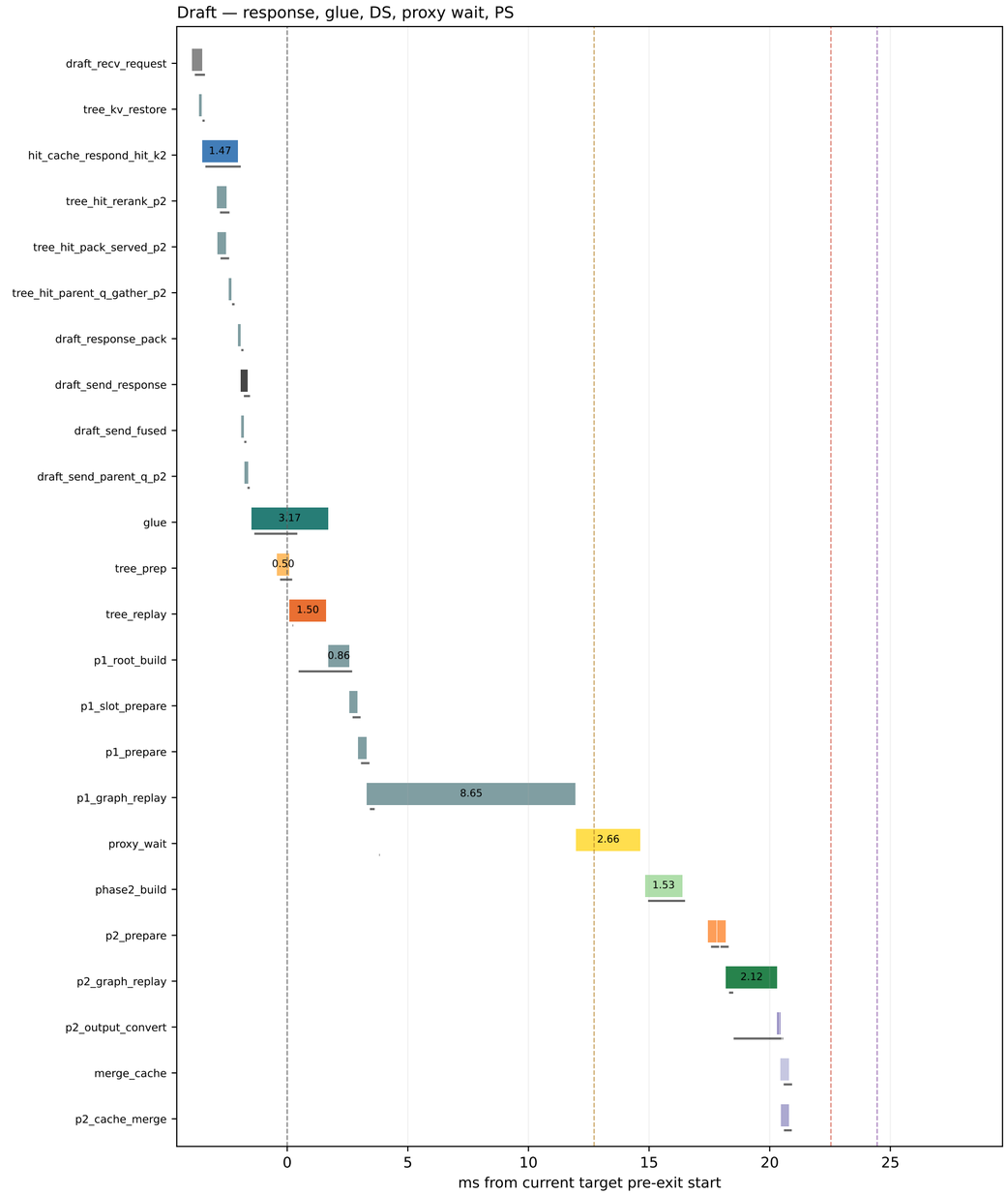
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
P2 hit / p75 target-step duration / request 2, UID 244_t0, step 101

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=3, AL incl. recovery=1; DS margin +2.659 ms (estimated); PS margin +1.731 ms; next outcome P2 hit





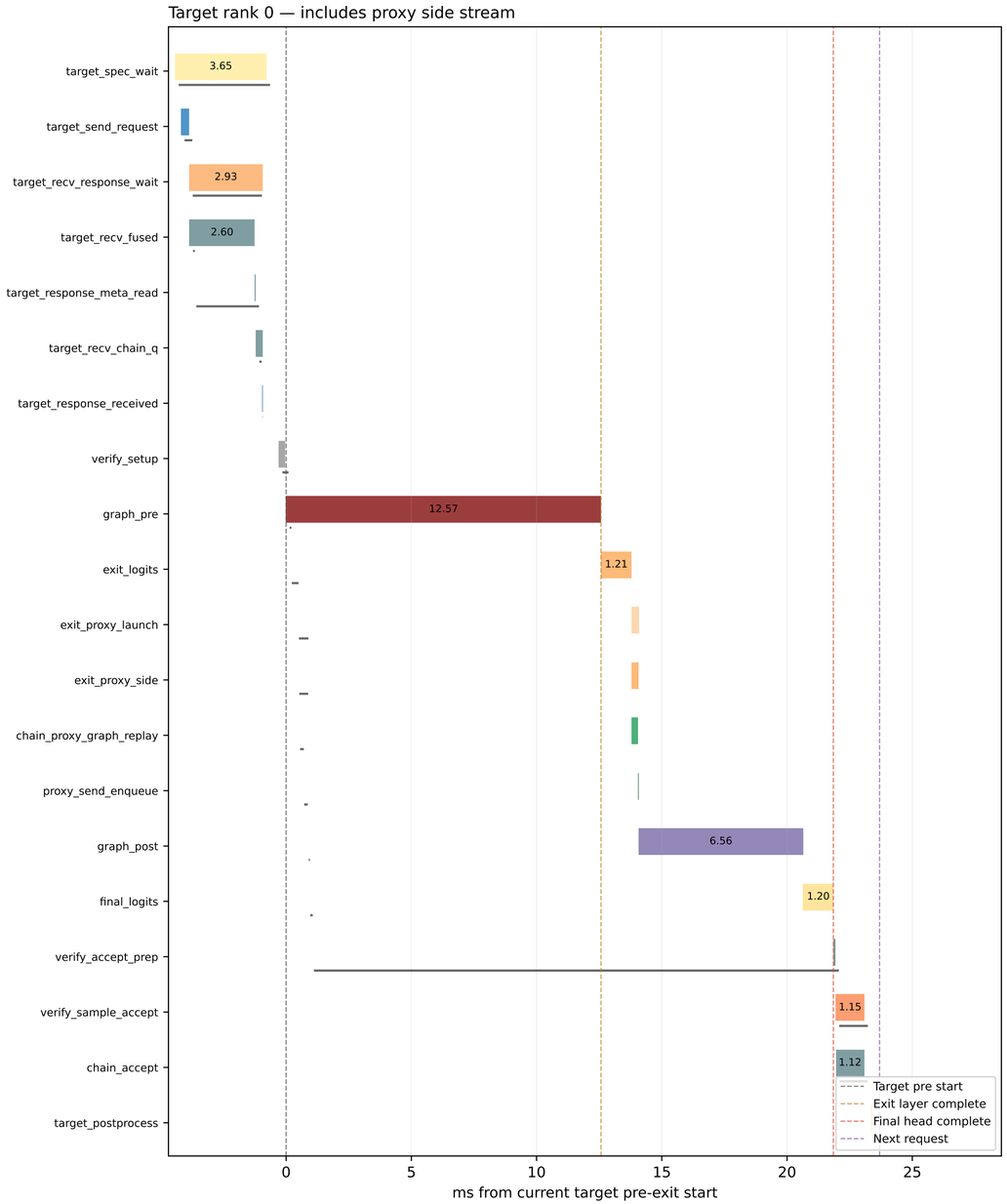
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

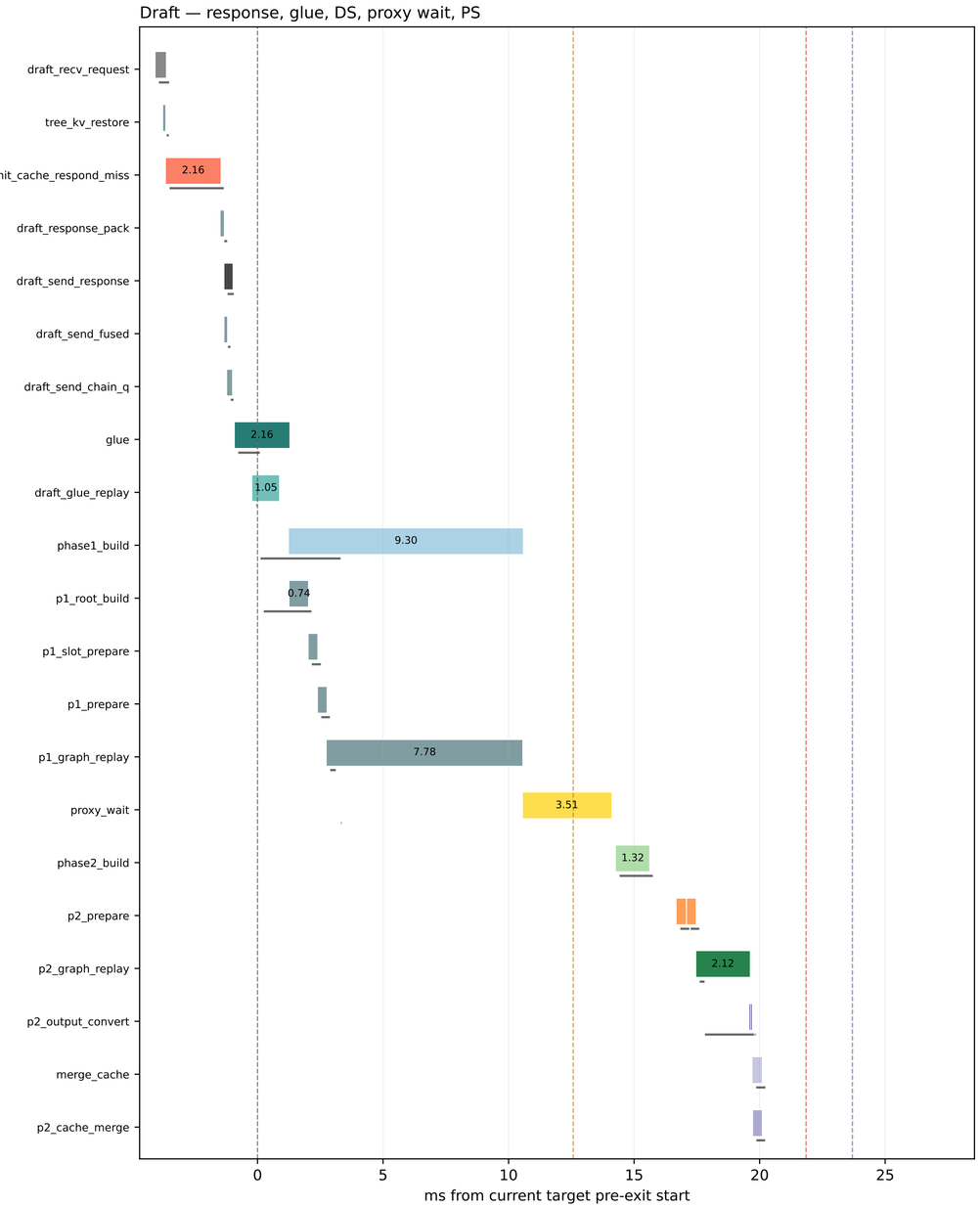
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Miss / p25 target-step duration / request 4, UID 288_t0, step 26

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=1, AL incl. recovery=1; DS margin +3.504 ms (estimated); PS margin +1.755 ms; next outcome P1 hit





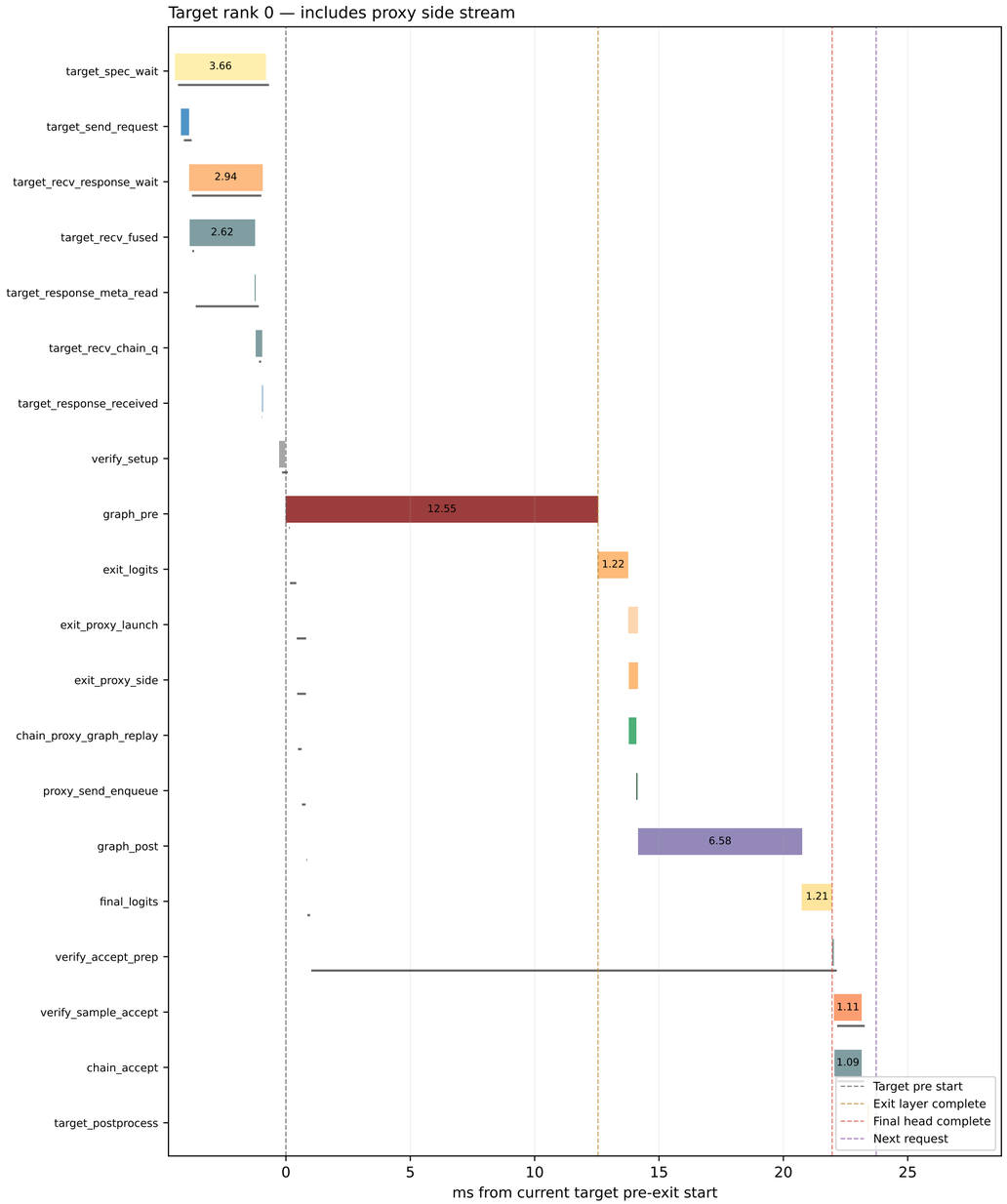
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

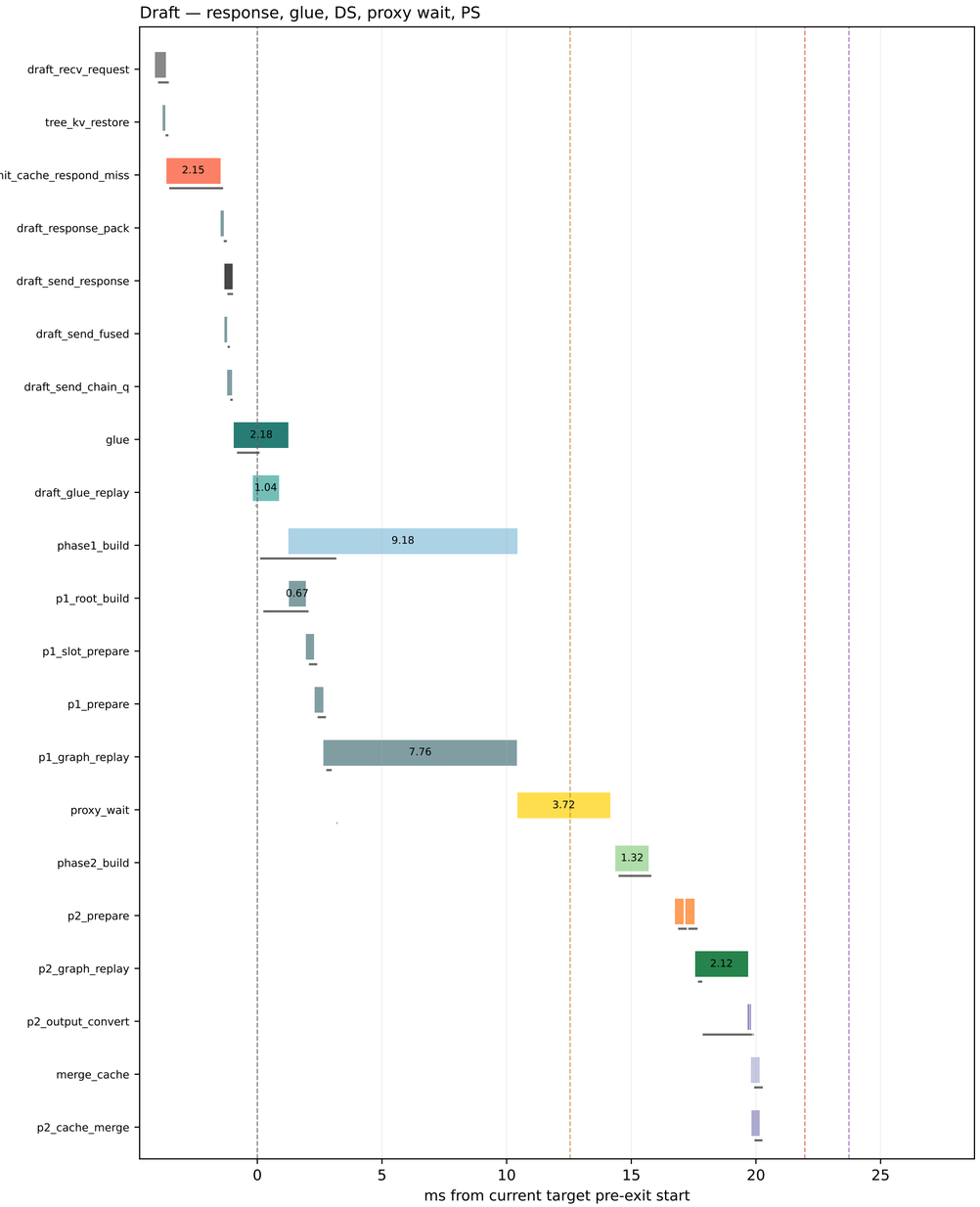
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Miss / p50 target-step duration / request 2, UID 244_t0, step 65

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=1, AL incl. recovery=1; DS margin +3.714 ms (estimated); PS margin +1.802 ms; next outcome P2 hit





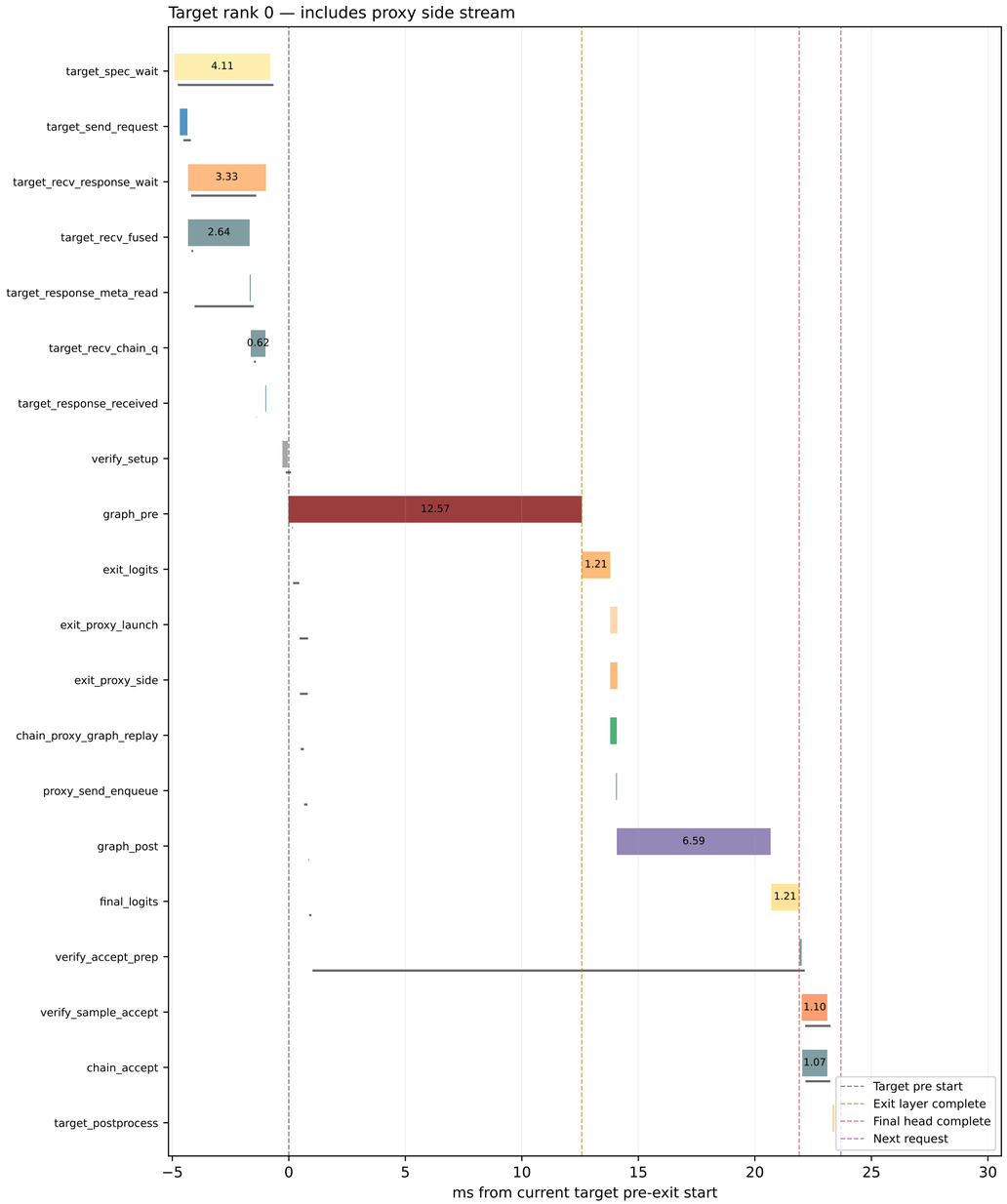
Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.

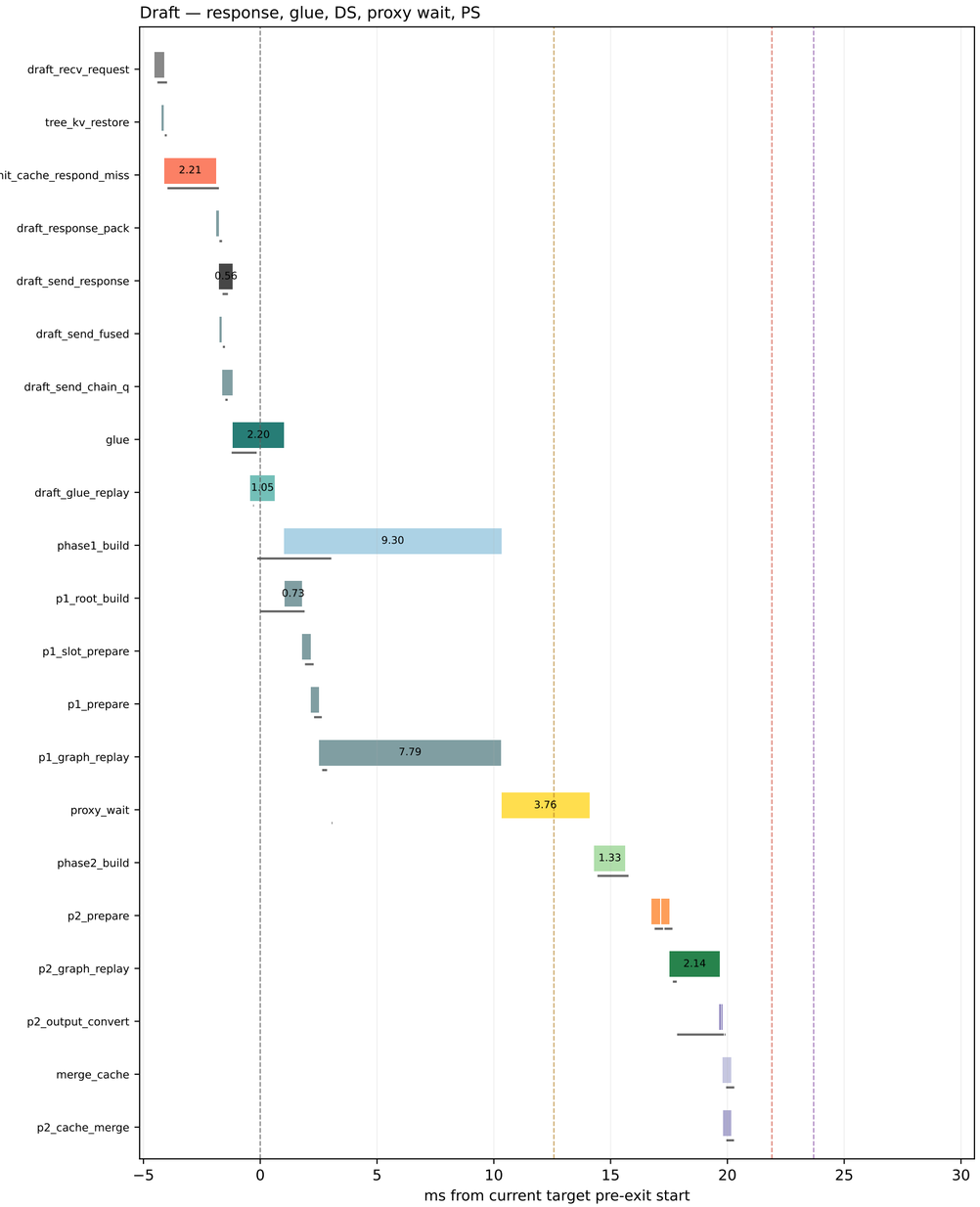
LayerSkip Llama3-8B + Axcel Llama3.1 drafter-340M
Miss / p75 target-step duration / request 4, UID 288_t0, step 15

exit 20 (0-based) | K1/K2 4/1 | G=M DS/PS 8/4 | C=3 | last | target GPUs=1 + draft GPU=1 | B=1, BF16, T=0.7

Measured v1.0 / 16dd4198 / eslab18 RTX3090 / 4 profiled turns, 387 eligible steps / output cap 256

valid proposal nodes=1, AL incl. recovery=2; DS margin +3.752 ms (estimated); PS margin +1.741 ms; next outcome P1 hit





Colored bars: raw CUDA-event elapsed spans. Black lines below: CPU dispatch. No clock shift / no minimum bar-width inflation. Nested rows are NOT additive.